Name: Dhatri. V CLASS: SEC: S6 ROLL No. SUB. R-programming

S.No.	Date	Name of the Lesson	Page No.	Teacher Sign Remarks
01.	3/2/26	learn all the basic of R-programming (Data types, variables, operators etc.		
02.	10/2/26	Operators & Decision Making in R.		
03.	17/2/26	learn the basic of functions in R and implement with examples		
04.	25/2/26	Implement data frame in R. Write a program to join colums and rows in a data frame.		
05.	10/3/26	Implement different String manipulation functions in R.		
06.	17/3/26	Implement different data structure in R (vectors, lists, data frame)		
07	24/3/26	Write a program to read a CSV file and analyze the data in the following colums.		
08.	31/3/26	create pie charts and bar charts using R.		
09.	31/3/26	Create a dataset & do statistical analysis on the data using R.		

1.1:

```
> roll_no <- 102 # Integer
> name <- "Ashish" # Character
> cgpa <- 9.45 # Numeric
> pass <- TRUE # Logical
> roll_no
[1] 102
> name
[1] "Ashish"
> cgpa
[1] 9.45
> pass
[1] TRUE
```

1.2

```
> emp_id <- 5002
> salary <- 45000.76
> department <- "IT"
> is_permanent <- TRUE
> emp_id
[1] 5002
> salary
[1] 45000.76
> department
[1] "IT"
> is_permanent
[1] TRUE
>
```

1.3

```
> product_name <- "Laptop"
> price <-55008
> stock_available <-26
> discount_eligible <- FALSE
> product_name
[1] "Laptop"
> price
[1] 55008
> stock_available
[1] 26
> discount_eligible
[1] FALSE
>
```

```
> city <- "China"
> temperature <- 36.5
> rain <- FALSE
> wind_speed <- 12.8
> city
[1] "China"
> temperature
[1] 36.5
> rain
[1] FALSE
> wind_speed
[1] 12.8
>
```

1.5

```
subject <- "science"
total_marks <- 100
marks_obtained <- 78
result <- marks_obtained >= 40
subject
] "science"
total_marks
] 100
marks_obtained
] 78
result
] TRUE
```

1.6

```
account_no <- 123456789
holder_name <- "raj kumar"
balance <- 25000.50
account_active <- TRUE
account_no
] 123456789
holder_name
] "raj kumar"
balance
] 25000.5
account_active
] TRUE
```

```
movie_title <- "Inception"
rating <- 9.8
reviews <- 1300
available <- TRUE
movie_title
] "Inception"
rating
] 9.8
reviews
] 1300
available
] TRUE
```

1.8

```
vehicle_count <- 351
location <- "Highway Road"
avg_speed <- 47.6
congested <- TRUE
vehicle_count
] 351
location
] "Highway Road"
avg_speed
] 47.6
congested
] TRUE
```

2.1

```
attendance <- 78
if(attendance >= 75){
  print("Eligible for Exam")
} else {
  print("Not Eligible for Exam")
}
] "Eligible for Exam"
```

2.2

```
age <- 25
income <- 35000
if(age > 21 & income > 30000){
  print("Eligible for Loan")
} else {
  print("Not Eligible for Loan")
}
1] "Eligible for Loan"
```

```
amount <- 4200
premium_member <- TRUE
if(amount > 5000 | premium_member){
  print("Discount Applied")
} else {
  print("No Discount")
}
1] "Discount Applied"
```

2.4

```
units <- 250
if(units < 100){
  print("Low Usage")
} else if(units <= 300){
  print("Medium Usage")
} else {
  print("High Usage")
}
1] "Medium Usage"
```

2.5

```
experience <- 6
rating <- 4.5
if(experience > 5 & rating > 4){
  print("Bonus Eligible")
} else {
  print("Not Eligible for Bonus")
}
L] "Bonus Eligible"
```

2.6

```
password <- "secure123"
if(nchar(password) >= 8){
  print("Valid Password")
} else {
  print("Invalid Password")
}
] "Valid Password"
```

2.7

```
temperature <- 45
if(temperature < 0 | temperature > 40){
  print("Extreme Temperature Warning")
} else {
  print("Normal Temperature")
}
.] "Extreme Temperature Warning"
```

```
marks <- 82
if(marks >= 90){
  grade <- "A"
} else if(marks >= 75){
  grade <- "B"
} else if(marks >= 50){
  grade <- "C"
} else {
  grade <- "Fail"
}
print(grade)
] "B"
```

2.9

2.10

```
bp <- 150
sugar <- 210
if(bp > 140 & sugar > 200){
  risk <- "High Risk"
} else if((bp >= 120 & bp <= 140) | (sugar >=
40 & sugar <= 200)){
  risk <- "Medium Risk"
} else {
  risk <- "Low Risk"
}
print(risk)
L] "High Risk"
```

```
> vehicle_count <- 180
> if(vehicle_count > 200){
+   print("Long Green Signal")
+ } else if(vehicle_count >= 100){
+   print("Medium Green Signal")
+ } else {
+   print("Short Green Signal")
+ }
[1] "Medium Green Signal"
> |
```

2.12

```
attendance <- 92
performance <- 8.5
project_completed <- TRUE
if(attendance >= 90 & performance >= 8 & proj
ct_completed){
  print("Eligible for Appraisal")
} else {
  print("Not Eligible for Appraisal")
}
1] "Eligible for Appraisal"
```

2.13

```
face_detected <- TRUE
multiple_faces <- FALSE
internet_connected <- TRUE
if(!face_detected | multiple_faces | !interne
_connected){
  print("Exam Disqualified")
} else {
  print("Exam Allowed")
}
L] "Exam Allowed"
```

2.14

```
 high_demand <- TRUE
available_drivers <- 30
if(high_demand & available_drivers < 50){
  print("Surge Pricing Applied")
} else {
  print("Normal Pricing")
}
] "Surge Pricing Applied"
```

2.15

```
> temperature <- 75
> pressure <- 130
> vibration <- 6
> if(temperature > 80 | pressure > 120 | vibr
ion > 7){
+   print("Machine Shutdown")
+ } else {
+   print("Machine Running Normally")
+ }
[1] "Machine Shutdown"
>
```

2.16

```
 entrance_score <- 78
twelfth_marks <- 82
age <- 22
if(entrance_score >= 70 & twelfth_marks >= 75
age <= 25){
  print("Admission Granted")
} else {
  print("Admission Rejected")
}
] "Admission Granted"
```

3.1

```
 simple_interest <- function(p, r, t){
    si <- (p * r * t) / 100
    return(si)
  }
simple_interest(10000, 5, 2)
] 1000
```

3.2

```
  check_result <- function(marks){
    if(marks >= 40){
      return("Pass")
    } else {
      return("Fail")
    }
  }
 check_result(65)
1] "Pass"
```

3.3

```
net_salary <- function(basic_salary){
    tax <- basic_salary * 0.10
    net <- basic_salary - tax
    return(net)
  }
net_salary(30000)
L] 27000
```

3.4

```
electricity_bill <- function(units){
    if(units <= 100){
      bill <- units * 2
    } else if(units <= 200){
      bill <- (100 * 2) + (units - 100) * 3
    } else {
      bill <- (100 * 2) + (100 * 3) + (units
200) * 5
    }
    return(bill)
  }
electricity_bill(250)
] 750
```

3.5

```
final_amount <- function(amount){

    if(amount >= 5000){     discount <- amount
0.20
     } else if(amount >= 2000){
       discount <- amount * 0.10
     } else {
       discount <- 0
     }
     return(amount - discount)
   }
final_amount(4500)
] 4050
```

3.6

```
celsius_to_fahrenheit <- function(c){
    f <- (c * 9/5) + 32
    return(f)
  }
celsius_to_fahrenheit(37)
] 98.6
```

3.7

```
  assign_grade <- function(marks){
    if(marks >= 90){
      "A"
    } else if(marks >= 75){
      "B"
    } else if(marks >= 50){
      "C"
    } else {
      "Fail"
    }
  }
assign_grade(88)
] "B"
```

3.8

```
    check_voting <- function(age){
       if(age >= 18){
         return("Eligible to Vote")
       } else {
         return("Not Eligible to Vote")
       }
     }
   check_voting(17)
   ] "Not Eligible to Vote"
```

3.9

```
  calculate_emi <- function(principal, rate, y
ars){
     monthly_rate <- rate / (12 * 100)
     months <- years * 12
     emi <- (principal * monthly_rate * (1 + m
nthly_rate)^months) /
       ((1 + monthly_rate)^months - 1)
     return(round(emi, 2))
   }
 calculate_emi(500000, 7.5, 10)
1] 5935.09
```

3.10

```
  hospital_bill <- function(consultation, room
charge, days){
     subtotal <- consultation + (room_charge *
ays)
     tax <- subtotal * 0.05
     total <- subtotal + tax
     return(total)
   }
 hospital_bill(500, 2000, 3)
1] 6825
```

3.12

```
    if(cart_value >= 3000){
      discount <- cart_value * 0.10
    } else {
      discount <- 0
    }
    if(cart_value >= 5000){
    delivery <- 0
    } else {
    delivery <- 100
    }
    final_amount <- cart_value - discount + delivery
    return(final_amount)
  }
order_total(4200)
1] 3880
```

3.12

```
calculate_gpa <- function(marks){
    avg <- mean(marks)
    if(avg >= 85){
      result <- "Distinction"
    } else if(avg >= 70){
      result <- "First Class"
    } else if(avg >= 50){
      result <- "Second Class"
    } else {
      result <- "Fail"
    }
    return(result)
  }
```

3.13

```
+     total_units <- peak_units + offpeak_uni
+     if(total_units > 300){
+       bill <- bill * 1.10
+     }
+     return(bill)
+   }
+ smart_meter_bill(180, 150)
+
```

```
insurance_premium(5000, 55, TRUE)
 insurance_premium <- function(base, age, sm
r){
    premium <- base
    if(age > 50){
      premium <- premium * 1.20
    }
    if(smoker){
      premium <- premium * 1.30
    }
    return(premium)
  }
insurance_premium(5000, 55, TRUE)
```

3.15

```
payroll <- function(basic){
    hra <- basic * 0.20
    da <- basic * 0.10
    gross <- basic + hra + da
    tax <- gross * 0.12
    net <- gross - tax
    return(net)
  }
payroll(40000)
```

3.16

```
+ water_usage_status <- function(litres){
+     if(litres <= 500){
+       "Normal"
+     } else if(litres <= 1000){
+       "High Usage"
+     } else {
+
+       "Critical"
+     }
+   }
+ water_usage_status(1200)
+
```

```
# Join columns
student_data <- cbind(students, marks)
student_data
# Add new student records
new_students <- data.frame(
  RollNo = c(104),
  Name = c("Anjali surya"),
  Dept = c("CSE"),
  Marks = c(90)
)
final_data <- rbind(student_data, new_students)
final_data
```

4.2

```
+ # Join columns
+ employees <- cbind(employee_details, salary_d
etails)
+ employees
+ # Add new employee
+ new_employee <- data.frame(
+   EmpID = 4,
+   Name = "Neha sri",
+   Dept = "IT",
+   Salary = 50000
+ )
+ employees <- rbind(employees, new_employee)
+ employees
+
```

4.3

```
# Add new patient
new_patient <- data.frame(
  PatientID = 203,
  Name = "Arjun",
  Age = 50,
  Cost = 20000
)
patient_records <- rbind(patient_records, new
atient)
patient_records
PatientID           Name Age  Cost
      201 Suresh lakshmi  45 12000
      202  Lakshmi priya  38 15000
      203          Arjun  50 20000
```

4.4

```
# Add new product
new_product <- data.frame(
  ProductID = 4,
  ProductName = "Bottle",
  Quantity = 75
)
inventory <- rbind(inventory, new_product)
inventory
ProductID ProductName Quantity
        1         Pen      100
        2        Book       50
        3         Bag       30
        4      Bottle       75
```

4.5

```
# Combine all columns
student_records <- cbind(students, marks, att
endance)
student_records
  RollNo       Name Dept Sem1 Sem2
      1     Arunja  CSE   78   80
      2     Binaya  ECE   82   85
      3 Charan sri   IT   90   88
  Attendance
          92
          88
          95
```

4.6

```
payroll
EmpID     Name    Dept Basic  HRA   DA
  101   Rahula      IT 30000 6000 3000
  102 Snehitha      HR 35000 7000 3500
  103    Kiran Finance 40000 8000 4000
Rating
   4.5
   4.2
   4.8
```

4.7

```
hospital_data <- rbind(hospital_data, new_pat
ent)
hospital_data
PatientID    Name Age      Disease Amount
      201  Suresh  45     Diabetes  15000
      202 Lakshmi  38 Hypertension  12000
      203   Arjun  52      Cardiac  25000
```

4.8

```
# Add new product
new_product <- data.frame(
  ProductID = 4,

  ProductName = "Smart Watch",
  UnitsSold = 90,
  Revenue = 900000
)
sales_data <- rbind(sales_data, new_product)
sales_data
ProductID ProductName UnitsSold Revenue
        1      Mobile       120 1200000
        2      Laptop        80 4000000
        3      Tablet        60 1800000
        4 Smart Watch        90  900000
```

4.9

```
agri_data <- rbind(agri_data, new_field)
agri_data
FieldID  Crop Moisture Temp ExpectedYield
      1  Rice       45   32           4.5
      2 Wheat       38   29           3.8
      3 Maize       40   30           4.1
```

4.10

```
> # Merge with attendance
> final_records <- merge(student_data, attendan
ce, by = "RollNo")
> final_records
  RollNo   Name Dept Sem1 Sem2 Attendance
1      1   Arun  CSE   78   80         92
2      2   Bina  ECE   82   85         88
3      3 Charan   IT   90   88         95
>
```

4.11

```
# Merge with ratings
payroll <- merge(payroll, rating, by = "EmpI
', all.x = TRUE)
payroll
EmpID  Name    Dept Basic  HRA   DA
  101 Rahul      IT 30000 6000 3000
  102 Sneha      HR 35000 7000 3500
  103 Kiran Finance    NA   NA   NA
Rating
   4.5
    NA
   4.8
```

4.12

```
# Merge patients with diagnosis
hospital_data <- merge(patients, diagnosis, b
= "PatientID", all.x = TRUE)
# Merge with billing
hospital_data <- merge(hospital_data, billin
 by = "PatientID")
hospital_data
PatientID    Name Age  Disease Amount
      201  Suresh  45 Diabetes  15000
      202 Lakshmi  38     <NA>  12000
      203   Arjun  52  Cardiac  25000
```

4.13

```
> sales_data <- merge(sales_data, revenue, by =
"ProductID", all = TRUE)
> sales_data
  ProductID ProductName UnitsSold Revenue
1         1      Mobile       120 1200000
2         2      Laptop        80 4000000
3         3      Tablet        60      NA
4         4 Smart Watch        NA  900000
> ✓ Demonstrates FULL OUTER JOIN
```

4.14

```
> # LEFT JOIN with yield
> agri_data <- merge(agri_data, yield, by = "Fi
eldID", all.x = TRUE)
> agri_data
  FieldID  Crop Moisture Temp ExpectedYield
1       1  Rice       45   32           4.5
2       2 Wheat       38   29            NA
3       3 Maize       NA   NA           4.1
>
```

5.1

```
> names <- c("aruna", "Balaji", "chitra")
> upper_names <- toupper(names)
> name_length <- nchar(upper_names)
> upper_names
[1] "ARUNA"  "BALAJI" "CHITRA"
> name_length
[1] 5 6 6
```

5.2

```
> emails <- c("aruna@gmail.com", "meenakshi@ya
oo.com", "ravikumar@outlook.com")
> usernames <- sub("@.*", "", emails)
> usernames
[1] "aruna"     "meenakshi" "ravikumar"
```

5.3

```
clean_input <- trimws(input)
clean_input
[1] "R Programming" "Data Science"
[3] "AI"
```

5.4

```
> product_codes <- c("PRD101", "PRD202", "ABC3
23")
> valid_codes <- startsWith(product_codes, "PR
")
> valid_codes
[1]  TRUE  TRUE FALSE
```

```
> report <- "error in file, error in system, er
ror occurred"
> updated_report <- gsub("error", "issue", repo
rt)
> updated_report
[1] "issue in file, issue in system, issue occu
rred"
```

5.6

```
> password <- "secure123"
> if(nchar(password) >= 9){
+   print("Strong Password")
+ } else {
+   print("Weak Password")
+ }
[1] "Strong Password"
```

5.7

```
word_count <- length(strsplit(sentence, " ")
[[1]])
word_count
[1] 6
```

5.8

```
> extensions <- sub(".*\\.", "", files)
> extensions
[1] "csv" "pdf" "png"
```

5.9

```
 names <- c("Gayathri sri", "Karthika", "Anit
a")
 abbr <- substr(names, 1, 3)
 abbr
1] "Gay" "Kar" "Ani"
```

5.10

```
 matched_names <- grep("a", names, value = TRU
)
 matched_names
1] "Aruna"     "Balaji"    "Ravikumar"
```

5.11

```
+   Name = c(" Gayathri sri  ", "RAVI kumar",
meenakshi"),
+   Email = c("Gayu@UNIV.edu", "ravi@univ.edu'
"meena@gmail.com"),
+   Department = c(" cse", "ECE ", " Cse ")
+ )
> students
             Name           Email
1  Gayathri sri     Gayu@UNIV.edu
2      RAVI kumar   ravi@univ.edu
3       meenakshi meena@gmail.com
  Department
1        cse
2       ECE
3       Cse
>
```

5.12

```
   products$Product <- tools::toTitleCase(tolo
er(products$Product))
   products$ProductCode <- sub("PRD-[0-9]+-",
', products$Code)
   products
     Product        Code ProductCode
Apple Phone PRD-101-APL         APL
 Samsung-Tv PRD-202-SAM         SAM
Dell Laptop PRD-303-DEL         DEL
```

5.13

```
  products$Product <- tools::toTitleCas
r(products$Product))
  products$ProductCode <- sub("PRD-[0-9
, products$Code)
  products
    Product        Code ProductCode
Apple Phone PRD-101-APL         APL
 Samsung-Tv PRD-202-SAM         SAM
Dell Laptop PRD-303-DEL         DEL
  products
    Product        Code ProductCode
Apple Phone PRD-101-APL         APL
 Samsung-Tv PRD-202-SAM         SAM
Dell Laptop PRD-303-DEL         DEL
```

6.1

```
total <- sum(marks)
average <- mean(marks)
highest <- max(marks)
total
.] 405
average
.] 81
highest
.] 98
```

6.2

```
min_temp <- min(temperature)
max_temp <- max(temperature)
min_temp
L] 59

max_temp
L] 98
```

6.3

```
> employee
$ID
[1] 101

$Name
[1] "Dhatri"

$Salary
[1] 1e+05

$Department
[1] "IT"

> employee$Name
[1] "Dhatri"
> employee$Salary
[1] 1e+05
```

6.4

```
> patient
$PatientID
[1] 202

$Name
[1] "Ashish"

$Age
[1] 23

$TestResults
[1] 120 140 130

> patient$TestResults
[1] 120 140 130
```

6.5

```
students <- data.frame(
  RollNo = c(1, 2, 3),
  Name = c("Ashish", "Dhatri", "srinu"),
Dept = c("CSE", "ECE", "IT"),
Marks = c(85, 78, 92)
)
students
RollNo   Name Dept Marks
     1 Ashish  CSE    85
     2 Dhatri  ECE    78
     3  srinu   IT    92
```

6.6

```
sales <- data.frame(
Month = c("Jan", "Aug", "Sep"),
Sales = c(50000, 62000, 58000)
)
total_sales <- sum(sales$Sales)
total_sales
] 170000
```

6.7

```
> prices
[1] 1200  850  560
> customer
$CustomerID
[1] 301

$Name
[1] "Anjali"

$City
[1] "Chennai"

> orders
     Item Price
1   Phone  1200
2 Headset   850
3 Charger   560
```

6.8

```
> student_profile
$Name
[1] "Arun"

$RollNo
[1] 105

$Marks
[1] 78 85 88

> class_results
  RollNo Total
     101   240
     102   255
     103   270
```

6.9:

```
student_profile
Name
1] "Arun"

RollNo
1] 105

Marks
1] 78 85 88

class_results
 RollNo Total
    101   240
    102   255
    103   270
```

6.10

```
scores <- c(85, 90, 88)
profile <- list(
Name = "Dhatri",
Age = 21,
Scores = scores

)
records <- data.frame(
Subject = c("Math", "CS", "Physics"),
Marks = scores
)
```

7:

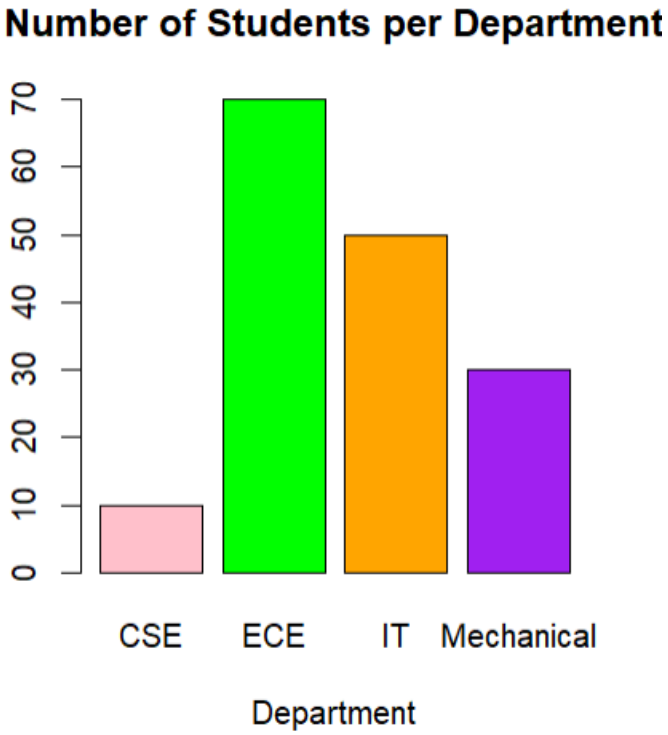
```
> # View first few rows
> head(sales)
        Date Product    Category Quantity Price
1 2026-01-01   Phone Electronics        5 20000
2 2026-01-01     Bag Accessories       10  1500
3 2026-01-02  Laptop Electronics        3 50000
4 2026-01-02 Headset Electronics        7  2500
5 2026-01-03     Bag Accessories        5  1500
6 2026-01-03   Phone Electronics        3 20000
> # Check structure and summary
> str(sales)
'data.frame':   6 obs. of  5 variables:
 $ Date    : chr  "2026-01-01" "2026-01-01" "2026-01-02" "2026-01-02" ...
 $ Product : chr  "Phone" "Bag" "Laptop" "Headset" ...
 $ Category: chr  "Electronics" "Accessories" "Electronics" "Electronics" ...
 $ Quantity: int  5 10 3 7 5 3
 $ Price   : int  20000 1500 50000 2500 1500 20000
> summary(sales)
     Date             Product
 Length:6           Length:6
 Class :character   Class :character
 Mode  :character   Mode  :character



   Category            Quantity        Price
 Length:6           Min.   : 3.0   Min.   : 1500
 Class :character   1st Qu.: 3.5   1st Qu.: 1750
 Mode  :character   Median : 5.0   Median :11250
```

```
 Mode  :character   Median : 5.0   Median :11250
                    Mean   : 5.5   Mean   :15917
                    3rd Qu.: 6.5   3rd Qu.:20000
                    Max.   :10.0   Max.   :50000
> # Remove extra spaces in Product and Category columns
> sales$Product <- trimws(sales$Product)
> sales$Category <- trimws(sales$Category)
> # Convert Date column to Date format
> sales$Date <- as.Date(sales$Date, format="%Y-%m-%d")
> # Add a column for total sales per row
> sales$TotalSales <- sales$Quantity * sales$Price
> # View cleaned data
> head(sales)
        Date Product    Category Quantity Price
1 2026-01-01   Phone Electronics        5 20000
2 2026-01-01     Bag Accessories       10  1500
3 2026-01-02  Laptop Electronics        3 50000
4 2026-01-02 Headset Electronics        7  2500
5 2026-01-03     Bag Accessories        5  1500
6 2026-01-03   Phone Electronics        3 20000
  TotalSales
1     100000
2      15000
3     150000
4      17500
5       7500
6      60000
> # Calculate total sales for each product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, sum)
>
> total_sales_product
```

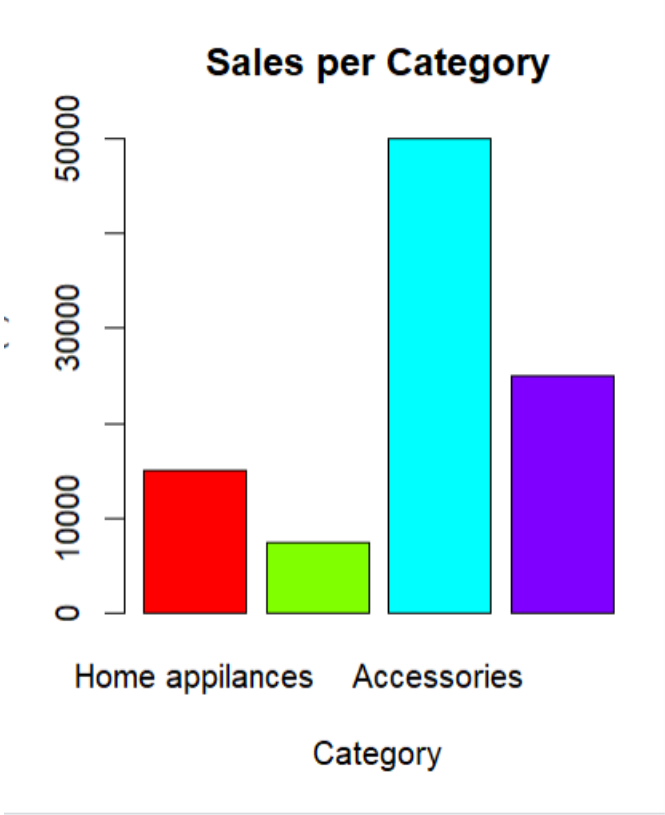

```
> # Calculate total sales for each product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, sum)
>
> total_sales_product
```


```
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
> top_product <- total_sales_product[which.max(total_sales_product$TotalSales), ]
> top_product
  Product TotalSales
4   Phone     160000
> quantity_per_category <- aggregate(Quantity ~ Category, data=sales, sum)
> quantity_per_category
     Category Quantity
1 Accessories       15
2 Electronics       18
> # Products with total sales greater than 10000
> high_sales <- total_sales_product[total_sales_product$TotalSales > 10000, ]
> high_sales
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
```

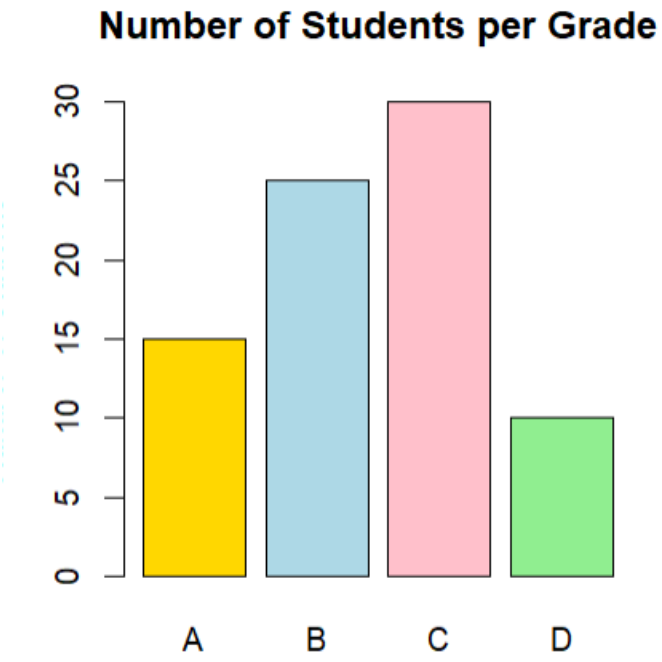
8.1


Number of Students per Department
0
10
20
30
40
50
60
70
CSE
ECE
IT
Mechanical
Department

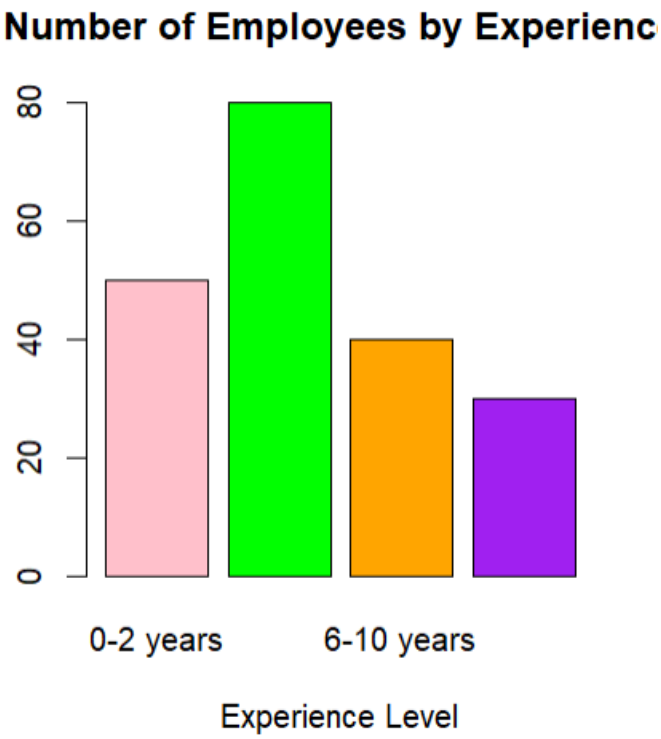
8.2


Sales per Category
0
10000
30000
50000
Home appilances
Accessories
Category

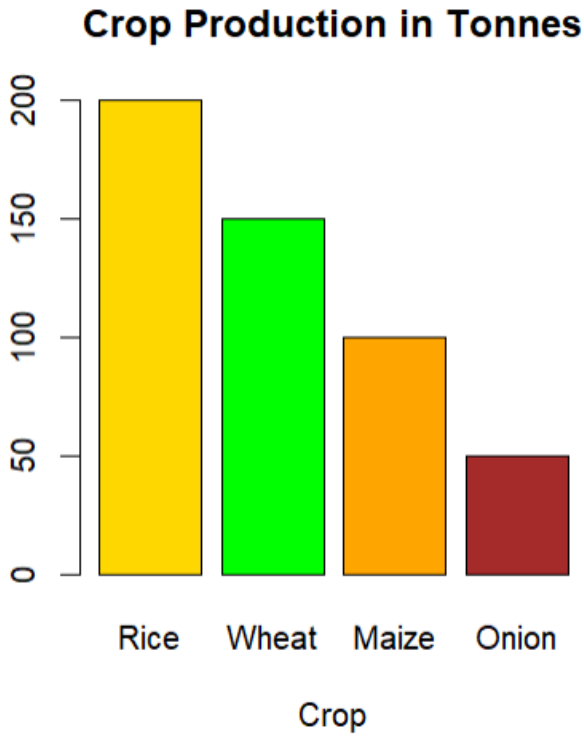
8.3


Number of Students per Grade
0
5
10
15
20
25
30
A
B
C
D

8.4


Number of Employees by Experienc
0
20
40
60
80
0-2 years
6-10 years
Experience Level

8.5



9.1:

```
> sd_marks
   Maths  Science  English
9.302927 9.809293 7.183314
> top_student
          Name Maths Science English Total
4 Anitha reddy    95      92      98   285
> above_avg_math
           Name Maths Science English Total
1        Arjuna    85      78      92   255
2     Meenakshi    90      88      80   258
4  Anitha reddy    95      92      98   285
6          dhat    88      80      85   253
8        chandu    92      90      91   273
10      Lakshmi    86      82      87   255
>
```

9.2

```
> # Patients older than 50
> older_patients <- patients[patients$Age > 50, ]
> mean_age
[1] 45.4
> median_age
[1] 46.5
> range_age
[1] 29 60
> sd_age
[1] 9.547542
> older_patients
  PatientID  Name Age
3       203  Ashi  52
5       205 srinu  60
7       207 Kiran  55
```

9.3

3

```
# Step 5: Product with max average sales
max_avg_product <- sales_stats[which.max(sales_stats$Mea
sales_stats
 Product Mean Max Min Variance
   Phone  5.4   7   4      1.3
  Laptop  3.4   5   2      1.3
 Headset  6.6   8   5      1.3
     Bag 10.0  12   8      2.5
  Camera  2.2   3   1      0.7
   Mouse  7.6   9   6      1.3
Keyboard  6.4   8   5      1.3
max_avg_product
Product Mean Max Min Variance
    Bag   10  12   8      2.5
```

9.4

```
> # Patients older than 50
> older_patients <- patients[patients$Age > 50, ]
> mean_age
[1] 45.4
> median_age
[1] 46.5
> range_age
[1] 29 60
> sd_age
[1] 9.547542
> older_patients
  PatientID  Name Age
3       203  Ashi  52
5       205 srinu  60
7       207 Kiran  55
>
```

9.5

```
overall_avg <- mean(employees$Salary)
high_earners <- employees[employees$Salary > overall_avg, ]
dept_stats
   Dept Salary.Mean Salary.Median
Finance    42333.33         42000
     HR    36000.00         36000
     IT    31500.00         31500
high_earners
EmpID     Name    Dept Salary
  103    Kiran Finance  40000
  106 Gayathri Finance  42000
  108  Lakshmi      HR  37000
  109    Arjun Finance  45000

# Products with rating > 4
top_rated <- ratings[ratings$Rating > 4, ]
mean_rating
1] 4.1
median_rating
1] 4
mode_rating
1] 4
var_rating
1] 0.5444444
top_rated
ProductID Product Rating
        2  Laptop      5
        5  Camera      5
        8 Charger      5
```